

Compactness and simplicity for the planar disk log gas

Abstract

For every fixed inverse temperature $\beta > 0$, the unscaled counting measures of the logarithmic gas in the open unit disk have tight laws. Every accumulation point, even when convergence is taken in the space of all nonnegative Radon measures, is a locally finite simple point process. We prove a uniform Gaussian upper tail for compact counts, derive a local two-point repulsion estimate, and show explicitly why these bounds exclude both diffuse mass and multiple atoms in the limit. All estimates specific to the model are proved in this note.

1 Model, topology, and statement

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$, let $B_r = \{z : |z| < r\}$, and let dA denote ordinary, unnormalized planar Lebesgue measure. Fix $\beta > 0$. For $N \geq 1$, the labelled particles have law

$$\mathbb{P}_{N,\beta}(dz_1 \cdots dz_N) = \frac{1}{Z_{N,\beta}} |\Delta_N(z)|^\beta \prod_{i=1}^N dA(z_i), \quad z \in \mathbb{D}^N, \quad \Delta_N(z) = \prod_{1 \leq i < j \leq N} (z_i - z_j). \quad (1.1)$$

Here $Z_{N,\beta}$ is the integral of the numerator over $\mathbb{D}^N$, and empty products equal one. The partition function is finite because all pair distances are at most 2, and positive because the integrand is positive off the collision diagonals. In particular, this law is absolutely continuous and its particles are distinct almost surely. Write

$$\Xi_N = \sum_{i=1}^N \delta_{z_i}, \quad M_N(r) = \Xi_N(B_r).$$

There is no spatial rescaling as N increases.

The space $\mathcal{M}_+(\mathbb{D})$ consists of nonnegative Radon measures on $\mathbb{D}$: their mass on every compact subset of $\mathbb{D}$ is finite. It carries the *vague topology*, in which $\mu_n \rightarrow \mu$ means

$$\mu_n(f) \rightarrow \mu(f), \quad \mu(f) = \int f d\mu, \quad f \in C_c(\mathbb{D}).$$

Here $C_c(\mathbb{D})$ is the space of continuous functions with compact support in $\mathbb{D}$. The subspace $\mathcal{M}_p(\mathbb{D})$ consists of locally finite counting measures, allowing integer multiplicities at an atom. A *point process* is a random element of $\mathcal{M}_p(\mathbb{D})$; it is *simple* if the mass of each atom is one almost surely. Convergence in law below refers to probability laws on these spaces, not to convergence of the measures $N^{-1}\Xi_N$.

Vague convergence on the open disk permits particles to escape to the boundary. This is essential: on the closed disk the unnormalized total mass is N , so these laws could not be tight.

Theorem 1.1 (Compactness and simple limits). *Fix $\beta > 0$. The laws of $(\Xi_N)_{N \geq 1}$ are tight in $\mathcal{M}_+(\mathbb{D})$. Every sequence of particle numbers tending to infinity therefore has a subsequence along which Ξ_N converges in law. Every accumulation point is a locally finite simple point process.*

For $0 < r < 1$ and every integer $m \geq 2$,

$$\sup_{N \geq 1} \mathbb{P}\{M_N(r) \geq m\} \leq \min \left\{ 1, \exp \left(m - \frac{\beta}{4} m(m-1) \log \frac{1}{r} \right) \right\}. \quad (1.2)$$

The same bound holds for $M(r) = \Xi(B_r)$ for every limiting process Ξ . For every compact $K \subset \mathbb{D}$, $q > 0$, and $t \in \mathbb{R}$,

$$\sup_N \mathbb{E}[\Xi_N(K)^q] < \infty, \quad \sup_N \mathbb{E} e^{t\Xi_N(K)} < \infty,$$

and these moments are finite for every limit. Along any subsequence $\Xi_{N_j} \Rightarrow \Xi$, the counts $M_{N_j}(r)$ converge in law to $M(r)$ for each $r < 1$, and

$$\mathbb{E} M_{N_j}(r)^q \longrightarrow \mathbb{E} M(r)^q, \quad \mathbb{E} e^{tM_{N_j}(r)} \longrightarrow \mathbb{E} e^{tM(r)}. \quad (1.3)$$

The proof has two separate parts. A radial comparison bounds the number of particles in every compact window and gives tightness. A two-point estimate then prevents different particles from merging in the limit. Simplicity does not follow from tightness alone: for example, $\delta_0 + \delta_{1/n} \rightarrow 2\delta_0$ vaguely, although each measure in the sequence is simple.

2 Radial comparison and compact-count tails

We first prove the two elementary inequalities used in the comparison.

Lemma 2.1 (Subharmonic means). *If P is holomorphic and $\beta > 0$, then $|P|^\beta$ is subharmonic. Its circular mean about any center is nondecreasing in the radius, for circles whose enclosed disks lie in its domain. More generally, for a nonnegative continuous subharmonic function u ,*

$$u(z) \leq \frac{1}{\pi(b^2 - a^2)} \int_{a < |w-z| < b} u(w) dA(w), \quad 0 \leq a < b, \quad (2.1)$$

provided the closed disk of radius b about z lies in the domain.

Proof. Put $\alpha = \beta/2$. Direct differentiation gives, for $\varepsilon > 0$,

$$\Delta(|P|^2 + \varepsilon)^\alpha = 4\alpha(|P|^2 + \varepsilon)^{\alpha-2}(\varepsilon + \alpha|P|^2)|P'|^2 \geq 0.$$

The locally uniform limit as $\varepsilon \downarrow 0$ is subharmonic. For smooth u , the derivative of the circular mean at radius s is $(2\pi s)^{-1} \int_{B(z,s)} \Delta u dA \geq 0$. Regularization proves the corresponding statement for continuous subharmonic functions. Integrate the circular submean inequality against $2\pi s ds$ over $a < s < b$ to obtain (2.1). $\square$

Lemma 2.2 (Increasing tilts of product measures). *Let μ be a product probability measure on finitely many intervals. If $F \geq 0$ is bounded, nondecreasing in each coordinate, and $\mu(F) > 0$, then the probability measure $\nu = F\mu/\mu(F)$ satisfies*

$$\nu(A) \leq \mu(A)$$

for every coordinatewise decreasing Borel set A .

Proof. For bounded increasing functions f, g of one variable and independent identically distributed X, X' , one has

$$2 \operatorname{Cov}(f(X), g(X)) = \mathbb{E}[(f(X) - f(X'))(g(X) - g(X'))] \geq 0.$$

For a product measure, induct on the number of coordinates. The conditional covariance given the last coordinate is nonnegative by induction. The conditional expectations are increasing functions of that last coordinate, so their covariance is also nonnegative. The covariance decomposition proves the same inequality in every dimension. Apply it to $f = F$ and $g = -\mathbf{1}_A$ to obtain $\mu(F\mathbf{1}_A) \leq \mu(F)\mu(A)$. $\square$

By absolute continuity, zero or repeated radii have probability zero. Sort the radii as $0 < r_1 < \dots < r_N < 1$ and relabel the angles accordingly. Sorting contributes only a constant $N!$ to the density. Define the ratios

$$t_k = \frac{r_k}{r_{k+1}}, \quad r_{N+1} = 1, \quad r_j = \prod_{k=j}^N t_k, \quad a_k = 2k + \frac{\beta}{2}k(k-1). \quad (2.2)$$

For $i < j$, factor

$$|z_i - z_j| = r_j \left| 1 - \left(\prod_{\ell=i}^{j-1} t_\ell \right) e^{i(\theta_i - \theta_j)} \right|.$$

The triangular Jacobian and the radial Vandermonde factors give

$$\prod_{j=1}^N r_j dr_j = \prod_{k=1}^N t_k^{2k-1} dt_k, \quad \prod_{j=1}^N r_j^{\beta(j-1)} = \prod_{k=1}^N t_k^{\beta k(k-1)/2}.$$

Indeed, the Jacobian of $t \mapsto r$ is $\prod_k t_k^{k-1}$ and $\prod_j r_j = \prod_k t_k^k$. After angular integration, the ratio density is proportional to

$$F_N(t) \prod_{k=1}^N t_k^{a_k-1}, \quad t \in (0, 1)^N, \quad (2.3)$$

where

$$F_N(t) = \int_{[0, 2\pi)^N} \prod_{i < j} \left| 1 - \left(\prod_{\ell=i}^{j-1} t_\ell \right) e^{i(\theta_i - \theta_j)} \right|^\beta \prod_{j=1}^N \frac{d\theta_j}{2\pi}.$$

This function is bounded, continuous, and nonnegative on $[0, 1]^N$.

Lemma 2.3. *The angular factor F_N is nondecreasing in every ratio t_k .*

Proof. Fix $k < N$ and all other ratios. Only the pairs $i \leq k < j$ depend on t_k ; their radial ratios have the form $t_k q_{ij}$. Average additionally over the common rotation $\theta_i \mapsto \theta_i + \phi$ for $i \leq k$. Invariance of angular Lebesgue measure leaves F_N unchanged. All noncrossing factors are unchanged, whereas the crossing product is

$$|P_\theta(t_k e^{i\phi})|^\beta, \quad P_\theta(w) = \prod_{i \leq k < j} (1 - q_{ij} e^{i(\theta_i - \theta_j)} w).$$

Its circular mean is nondecreasing in t_k by Lemma 2.1. Integration against the remaining nonnegative factors preserves this inequality. The factor F_N does not depend on t_N . $\square$

Let U_k , $1 \leq k \leq N$, be independent with density $a_k u^{a_k-1}$ on $(0, 1)$. Equation (2.3) is their product law tilted by F_N , whose mean is positive. For $N \geq m$, the event $M_N(r) \geq m$ is the decreasing event $\prod_{k=m}^N t_k < r$. Hence Lemma 2.2 gives

$$\mathbb{P}\{M_N(r) \geq m\} \leq \mathbb{P}\left\{ \sum_{k=m}^N E_k > \log \frac{1}{r} \right\}, \quad E_k = -\log U_k. \quad (2.4)$$

The variables E_k are independent exponentials with rates a_k : $\mathbb{P}(E_k > x) = e^{-a_k x}$ for $x \geq 0$.

For $m \geq 2$, take $\lambda = \beta m(m-1)/4$. For every $k \geq m$, $a_k \geq \beta k(k-1)/2$ and $\lambda/a_k \leq 1/2$. Using $-\log(1-x) \leq 2x$ for $0 \leq x \leq 1/2$,

$$\begin{aligned} \log \mathbb{E} \exp \left(\lambda \sum_{k=m}^N E_k \right) &= \sum_{k=m}^N -\log(1 - \lambda/a_k) \\ &\leq 2\lambda \sum_{k=m}^N a_k^{-1} \leq \frac{4\lambda}{\beta} \sum_{k=m}^{\infty} \frac{1}{k(k-1)} = m. \end{aligned}$$

Exponential Markov inequality and (2.4) prove (1.2) at finite N ; if $N < m$, its left side is zero. In particular, for every $r < 1$, $q > 0$, and $t > 0$,

$$\sup_N \mathbb{E} M_N(r)^q < \infty, \quad \sup_N \mathbb{E} e^{tM_N(r)} < \infty. \quad (2.5)$$

For example, the terms with $m \geq 2$ in the exponential-moment sum are bounded by $\exp((t+1)m - c_r m(m-1))$ with $c_r = (\beta/4) \log(1/r) > 0$, a summable sequence.

3 Tightness in the space of random measures

We recall the required topological facts and explain the deterministic compactness and closure arguments. For a locally compact, second countable space such as $\mathbb{D}$, $\mathcal{M}_+(\mathbb{D})$ with the vague topology is Polish, meaning that its topology can be given by a complete separable metric. We use Prokhorov's theorem: a family of probability laws on a Polish space is tight if, for every $\varepsilon > 0$, a compact set has probability at least $1 - \varepsilon$ under every law; such a family is relatively sequentially compact for convergence in law. For the Polish property and Prokhorov's theorem, see [1, Theorems A2.6.III and A2.4.I], respectively.

Lemma 3.1 (Local mass bounds and counting-measure closure). *A family in $\mathcal{M}_+(\mathbb{D})$ whose masses are uniformly bounded on each compact subset of $\mathbb{D}$ is relatively compact for the vague topology. Moreover, $\mathcal{M}_p(\mathbb{D})$ is closed in $\mathcal{M}_+(\mathbb{D})$.*

Proof. For the first assertion, choose a compact exhaustion and countable uniformly dense families of continuous test functions supported in its successive members. The integrals of each test function are bounded by the corresponding local mass bound. Diagonal extraction gives convergence for this countable family. Uniform approximation and the same mass bounds extend convergence to every function in $C_c(\mathbb{D})$. The limiting functional is positive and locally bounded, so the Riesz representation theorem identifies it with a Radon measure. Thus every sequence has a vaguely convergent subsequence.

For closure, suppose that counting measures ξ_n converge vaguely to $\mu \in \mathcal{M}_+(\mathbb{D})$. If a relatively compact ball B satisfies $\mu(\partial B) = 0$, approximation of its indicator by continuous functions gives $\xi_n(B) \rightarrow \mu(B)$. Therefore $\mu(B)$ is an integer. For every $x \in \mathbb{D}$ there are arbitrarily small balls centered at x with μ -null boundary: among radii below any fixed compactly contained radius, only countably many circles can have positive mass. Along decreasing such radii, $\mu(B(x, s)) \downarrow \mu(\{x\})$. These ball masses are nonnegative integers and bounded above, so they eventually stabilize. Thus $\mu(\{x\})$ is an integer, and if it is zero some neighborhood of x has zero mass. It follows that every point of the support of μ is an atom of positive integer mass. There are finitely many such atoms on each compact set because μ is Radon. Hence μ is a locally finite counting measure. $\square$

The local compactness criterion and the closure assertion also appear in [1, Corollary A2.6.V] and [2, Lemma 9.1.V], respectively.

Choose $r_\ell \uparrow 1$. Given $\varepsilon > 0$, the tail bound allows us to choose integers b_ℓ with

$$\sup_N \mathbb{P}\{\Xi_N(B_{r_\ell}) > b_\ell\} < \varepsilon 2^{-\ell}.$$

Consider the subset of the full measure space

$$\mathcal{K} = \bigcap_{\ell \geq 1} \{\mu \in \mathcal{M}_+(\mathbb{D}) : \mu(B_{r_\ell}) \leq b_\ell\}.$$

Mass on an open relatively compact set is lower semicontinuous for vague convergence. Thus $\mathcal{K}$ is closed. Every compact subset of $\mathbb{D}$ lies in some B_{r_ℓ} , so Lemma 3.1 makes $\mathcal{K}$ relatively compact, hence compact. A union bound gives $\inf_N \mathbb{P}(\Xi_N \in \mathcal{K}) \geq 1 - \varepsilon$. Prokhorov's theorem proves tightness in $\mathcal{M}_+(\mathbb{D})$.

Now fix any convergent subsequence $\Xi_{N_j} \Rightarrow \Xi$ in that space. Since $\mathcal{M}_p(\mathbb{D})$ is closed and contains every Ξ_N , Portmanteau's theorem shows that $\Xi \in \mathcal{M}_p(\mathbb{D})$ almost surely. This excludes diffuse mass in the limit; multiplicities still require proof.

The tail bound already passes to this counting-measure limit. The set $\{\mu : \mu(B_r) > m - 1/2\}$ is open by lower semicontinuity. Thus Portmanteau and the integer-valued counts give

$$\mathbb{P}\{\Xi(B_r) \geq m\} \leq \liminf_j \mathbb{P}\{\Xi_{N_j}(B_r) \geq m\},$$

which proves (1.2) for Ξ and all its local moment bounds. This argument requires no assumption about boundary points or simplicity.

4 A local two-point bound

The finite-system one- and two-point densities, relative to dA and $dA \otimes dA$, are defined by

$$\begin{aligned} \mathbb{E} \sum_i f(z_i) &= \int_{\mathbb{D}} f(z) \rho_{1,N}(z) dA(z), \\ \mathbb{E} \sum_{i \neq j} g(z_i, z_j) &= \int_{\mathbb{D}^2} g(z, w) \rho_{2,N}(z, w) dA(z) dA(w). \end{aligned}$$

The pairs in the second sum are ordered. Integration of (1.1) yields, for $p = 1, 2$ and $N \geq p$,

$$\rho_{p,N}(z) = |\Delta_p(z)|^\beta h_{p,N}(z), \quad (4.1)$$

where $(N)_p = N(N-1) \cdots (N-p+1)$ and

$$h_{p,N}(z) = \frac{(N)_p}{Z_{N,\beta}} \int_{\mathbb{D}^{N-p}} |\Delta_{N-p}(y)|^\beta \prod_{i=1}^p \prod_{\ell=1}^{N-p} |z_i - y_\ell|^\beta \prod_{\ell=1}^{N-p} dA(y_\ell). \quad (4.2)$$

For $N < p$ set these functions to zero. For each fixed N , they are continuous by dominated convergence. They are separately subharmonic in their arguments, since the integrands are powers of moduli of holomorphic polynomials, integrated against a positive measure independent of z .

Proposition 4.1 (Uniform local intensity and repulsion bounds). *For every $r < 1$ there is $C_{\beta,r} < \infty$, independent of N , such that, for $z, w \in \overline{B_r}$,*

$$\rho_{1,N}(z) \leq C_{\beta,r}, \quad \rho_{2,N}(z, w) \leq C_{\beta,r} |z - w|^\beta. \quad (4.3)$$

Proof. Fix $r < R < 1$ and put $\eta = (R - r)/8$. The disk submean inequality gives

$$h_{1,N}(z) \leq \frac{1}{\pi\eta^2} \int_{B_R} \rho_{1,N}(u) dA(u) = \frac{\mathbb{E}M_N(R)}{\pi\eta^2}.$$

This is uniformly bounded by (2.5).

For $h_{2,N}$ we average its two arguments over separated regions. If $|z - w| \geq 4\eta$, use the disks $B(z, \eta)$ and $B(w, \eta)$; points u, v in these disks satisfy $|u - v| \geq 2\eta$. If $|z - w| < 4\eta$, use $B(z, \eta)$ and the annulus $6\eta < |v - w| < 7\eta$. Then $|u - v| \geq \eta$. Both choices lie inside B_R ; each averaging region has area at least $\pi\eta^2$. Successive applications of Lemma 2.1 therefore give in either case

$$\begin{aligned} h_{2,N}(z, w) &\leq \frac{\eta^{-\beta}}{\pi^2\eta^4} \int_{B_R^2} |u - v|^\beta h_{2,N}(u, v) dA(u) dA(v) \\ &= \frac{\mathbb{E}[M_N(R)(M_N(R) - 1)]}{\pi^2\eta^{\beta+4}}. \end{aligned}$$

The second moment bound is uniform in N . Multiplication by $|z - w|^\beta$ proves the second assertion. $\square$

5 Why every accumulation point is simple

Let $\Xi_{N_j} \Rightarrow \Xi$ be the subsequence fixed above. We already know that Ξ is a locally finite counting measure, with multiplicities permitted. For such a measure ξ , its second *factorial measure* $\xi^{[2]}$ counts ordered pairs of distinct particle labels, including pairs at the same location if an atom has multiplicity. For $f \in C_c(\mathbb{D}^2)$,

$$\langle f, \xi^{[2]} \rangle = \int f(z, w) d\xi(z) d\xi(w) - \int f(z, z) d\xi(z). \quad (5.1)$$

For example, an atom of mass a contributes $a(a - 1)$ ordered pairs on the diagonal. This formula is valid before simplicity has been established.

Both terms on the right of (5.1) are continuous under vague convergence. For the product-measure term, approximate f uniformly on a compact product by finite sums of products of one-variable continuous functions, with compactly supported cutoffs. Vaguely convergent measures have uniformly bounded mass on each compact set, so this approximation reduces the assertion to the definition of vague convergence. The diagonal function $z \mapsto f(z, z)$ is itself continuous with compact support.

If the coordinate projections of the support of f lie in a compact set $K \subset B_S$, then

$$|\langle f, \Xi_{N_j}^{[2]} \rangle| \leq \|f\|_\infty M_{N_j}(S)^2.$$

The uniform fourth moment in (2.5) makes these random variables uniformly integrable. The continuous mapping theorem and uniform integrability imply

$$\mathbb{E}\langle f, \Xi_{N_j}^{[2]} \rangle \longrightarrow \mathbb{E}\langle f, \Xi^{[2]} \rangle. \quad (5.2)$$

The same argument, using the second moment bound, gives $\mathbb{E}\Xi_{N_j}(g) \rightarrow \mathbb{E}\Xi(g)$ for $g \in C_c(\mathbb{D})$.

Write $\alpha_1 = \mathbb{E}\Xi$ and $\alpha_2 = \mathbb{E}\Xi^{[2]}$ for the limiting intensity and second factorial moment measures. They are locally finite by the count moment bounds. For nonnegative continuous tests supported in B_r or B_r^2 , Proposition 4.1 and (5.2) show that

$$\alpha_1|_{B_r} \leq C_{\beta,r} dA, \quad \alpha_2|_{B_r^2} \leq C_{\beta,r} |z - w|^\beta dA(z) dA(w). \quad (5.3)$$

These are inequalities of Radon measures: domination on nonnegative continuous compactly supported tests extends to all Borel sets by regularity. In particular, α_2 gives no mass to the diagonal. For every $r < 1$,

$$0 = \alpha_2\{(z, z) : z \in B_r\} = \mathbb{E} \sum_{x \in B_r} \Xi(\{x\})(\Xi(\{x\}) - 1).$$

The sum is nonnegative. It is therefore zero almost surely, so every atom in B_r has mass at most one. Taking a countable exhaustion $r \uparrow 1$ proves that Ξ is simple almost surely. This completes the assertion about accumulation points in the full space of random measures.

The first inequality in (5.3) also proves that Ξ almost surely avoids every fixed area-null Borel subset of $\mathbb{D}$. In particular, for every fixed $r < 1$, $\Xi(\partial B_r) = 0$ almost surely. Counting B_r is continuous for vague convergence at any measure with this property: approximate its indicator from inside and outside by continuous functions. The continuous mapping theorem now gives $M_{N_j}(r) \Rightarrow M(r)$. Uniform bounds on higher powers and on $e^{sM_N(r)}$ for every $s > 0$ give uniform integrability of each fixed power and exponential. This proves (1.3) and finishes the proof of Theorem 1.1.

Remark 5.1 (Quantitative exclusion of close pairs). The preceding proof gives a uniform estimate for both the finite processes and every limit. If $K \subset B_r$ is compact and $\varepsilon > 0$, then

$$\mathbb{P}\{\text{two distinct points in } K \text{ have distance less than } \varepsilon\} \leq \frac{C_{\beta,r} \pi A(K)}{\beta + 2} \varepsilon^{\beta+2},$$

where $A(K)$ is the area of K . Indeed, such a pair contributes at least two to the ordered pair count. Integrating the two-point bound uses $\int_{|u| < \varepsilon} |u|^\beta dA(u) = 2\pi \varepsilon^{\beta+2}/(\beta + 2)$.

Remark 5.2 (The role of positive inverse temperature). At $\beta = 0$ the points are independent and uniform on $\mathbb{D}$. For $0 < r < 1$, $M_N(r)$ is binomial with parameters N and r^2 , so it tends to infinity in probability. The local tightness result therefore uses the assumption $\beta > 0$ essentially.

References

- [1] D. J. Daley and D. Vere-Jones, *An Introduction to the Theory of Point Processes*, vol. I: Elementary Theory and Methods, second edition, Springer, 2003. Appendix A2.4–A2.6.
- [2] D. J. Daley and D. Vere-Jones, *An Introduction to the Theory of Point Processes*, vol. II: General Theory and Structure, second edition, Springer, 2008. Section 9.1.